4.6.2 Summary

The LLS problem can be states as: Given $A \in \mathbb{C}^{m \times n}$ and $b \in \mathbb{C}^m$ find $\widehat{x} \in \mathbb{C}^n$ such that

$$\| b - A\widehat{x} \|_2 = \min_{x \in \mathbb{C}^n} \| b - Ax \|_2.$$

Given $A \in \mathbb{C}^{m \times n}$,

- The column space, $\mathcal{C}(A)$, which is equal to the set of all vectors that are linear combinations of the columns of A

  $$\{ y \mid y = Ax \}.$$

- The null space, $\mathcal{N}(A)$, which is equal to the set of all vectors that are mapped to the zero vector by A

  $$\{ x \mid Ax = 0 \}.$$

- The row space, $\mathcal{R}(A)$, which is equal to the set

  $$\{ y \mid y^H = x^H A \}.$$

  Notice that $\mathcal{R}(A) = \mathcal{C}(A^H)$.

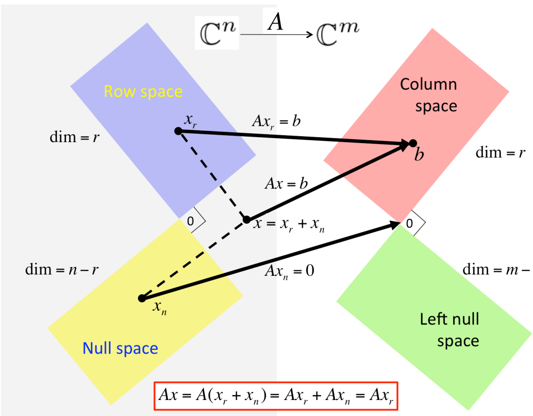
- The left null space, which is equal to the set of all vectors

  $$\{ x \mid x^H A = 0 \}.$$

  Notice that this set is equal to $\mathcal{N}(A^H)$.

- If $Ax = b$ then there exist $x_r \in \mathcal{R}(A)$ and $x = x_r + x_n$ where $x_r \in \mathcal{R}(A)$ and $x_n \in \mathcal{N}(A)$.

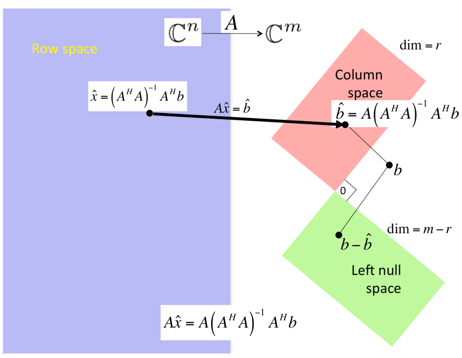
These insights are summarized in the following picture, which also captures the orthogonality of the spaces.



If A has linearly independent columns, then the solution of LLS, $\widehat{x}$, equals the solution of the normal equations

$$(A^H A)\widehat{x} = A^H b.$$

as summarized in



The (left) pseudo inverse of A is given by $A^\dagger = (A^H A)^{-1} A^H$ so that the solution of LLS is given by $\widehat{x} = A^\dagger b$.

Definition 4.6.2.1. Condition number of matrix with linearly independent columns. Let $A \in \mathbb{C}^{m \times n}$ have linearly independent columns (and hence $n \leq m$). Then its condition number (with respect to the 2-norm) is defined by

$$\kappa_2(A) = \|A\|_2 \|A^\dagger\|_2 = \frac{\sigma_0}{\sigma_{n-1}}.$$

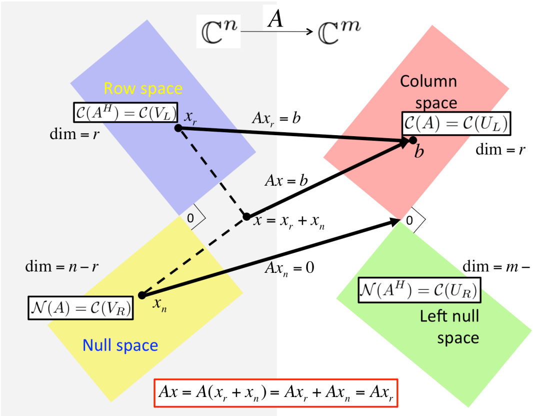
Assuming A has linearly independent columns, let $\hat{b} = A\widehat{x}$ where $\hat{b}$ is the projection of b onto the column space of A (in other words, $\widehat{x}$ solves the LLS problem), $\cos(\theta) = \|\hat{b}\|_2 / \|b\|_2$, and $\hat{b} + \delta\hat{b} = A(\widehat{x} + \delta\widehat{x})$, where $\delta\hat{b}$ equals the projection of δb onto the column space of A. Then

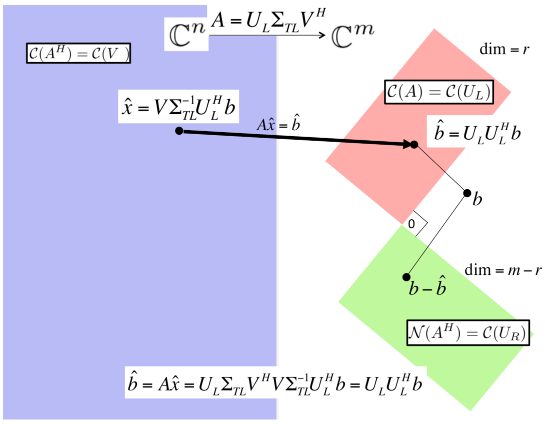
$$\frac{\|\delta\widehat{x}\|_2}{\|\widehat{x}\|_2} \leq \frac{1}{\cos(\theta)} \frac{\sigma_0}{\sigma_{n-1}} \frac{\|\delta\widehat{b}\|_2}{\|b\|_2}$$

captures the sensitivity of the LLS problem to changes in the right-hand side.

Theorem 4.6.2.2. *Given $A \in \mathbb{C}^{m\times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(U_L \mid U_R \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(V_L \mid V_R \right)^H$ its SVD. Then*

- $\mathcal{C}(A) = \mathcal{C}(U_L)$,
- $\mathcal{N}(A) = \mathcal{C}(V_R)$,
- $\mathcal{R}(A) = \mathcal{C}(A^H) = \mathcal{C}(V_L)$, *and*
- $\mathcal{N}(A^H) = \mathcal{C}(U_R)$.





If A has linearly independent columns and $A = U_L \Sigma_{TL} V_L^H$ is its Reduced SVD, then

$$\widehat{x} = V_L \Sigma_{TL}^{-1} U_L^H b$$

solves LLS.

Given $A \in \mathbb{C}^{m\times n}$, let $A = U_L \Sigma_{TL} V_L^H$ equal its Reduced SVD and $A = \left(U_L \mid U_R \right) \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \left(V_L \mid V_R \right)^H$ its SVD. Then

$$\widehat{x} = V_L \Sigma_{TL} U_L^H b + V_R z_b,$$

is the general solution to LLS, where z_b is any vector in $\mathbb{C}^{n-r}$.

Theorem 4.6.2.3. *Assume $A \in \mathbb{C}^{m\times n}$ has linearly independent columns and let $A = QR$ be its QR factorization with orthonormal matrix $Q \in \mathbb{C}^{m\times n}$ and upper triangular matrix $R \in \mathbb{C}^{n\times n}$. Then the LLS problem*

$$\text{Find } \widehat{x} \in \mathbb{C}^n \text{ such that } \|b - A\widehat{x}\|_2 = \min_{x\in\mathbb{C}^n} \|b - Ax\|_2$$

is solved by the unique solution of

$$R\widehat{x} = Q^H b.$$

Solving LLS via Gram-Schmidt QR factorization for $A \in \mathbb{C}^{m\times n}$:

- Compute QR factorization via (Classical or Modified) Gram-Schmidt: approximately $2mn^2$ flops.
- Compute $y = Q^H b$: approximately $2mn^2$ flops.
- Solve $R\widehat{x} = y$: approximately n^2 flops.

Solving LLS via Householder QR factorization for $A \in \mathbb{C}^{m\times n}$:

- Householder QR factorization: approximately $2mn^2 - \frac{2}{3}n^3$ flops.
- Compute $y_T = Q^H bnn$ by applying Householder transformations: approximately $4mn - 2n^2$ flops.
- Solve $R_{TL}\widehat{x} = y_T$: approximately n^2 flops.